بسم الله الرحمن الرحيم 24+17+7+39+10 = 88

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Matkul : Pengukuran Arus Umum

1. a) Diketahui : V/div : 0,5 mV
n : 3 kotak Vertikal ✓

Jwb : V_{PP} : n x V/div
: 3 x 0,5 mV atau 3 x (0,5 x 10^{-3})
: 1,5 mV atau 1,5 x 10^{-3}

$V_P = \frac{V_{PP}}{2} = \frac{1,5\ mV}{2} = 0,75\ mV$

$V_{RMS} = \frac{V_P}{\sqrt{2}} = \frac{0,75 \times 10^{-3}}{\sqrt{2}} = 5,303 \times 10^{-4} = 0,530\ mV$

b) Diketahui : t/div : 0,2 ms
m : 2 kotak Horizontal ✓

Jwb : $f = \frac{1}{m \times t/div} = \frac{1}{2 \times (0,2 \times 10^{-3})} = \frac{10^3}{2 \times 0,2} = \frac{1000}{0,4} = 2500\ Hz \approx 2,5\ kHz$ ✓

c) $V_{RMS} = \frac{V_P}{\sqrt{2}} = \frac{0,75 \times 10^{-3}}{\sqrt{2}} = 0,530\ mV$ ✓

2. a) Mode XY → Lissajous condong ke kiri, maka $\varphi = 180 - \sin^{-1}\left(\frac{2,5}{5}\right) = 150°$ ✓
Y_1 : 2,5 → di tengah antara 2 dan 3 ✓
Y_2 : 5 → ditengah antara 4 dan 6

b) Dual mode → $\varphi = \frac{a}{c} \times 180° = \frac{0,4}{2,5} \times 180° = 28,8°$

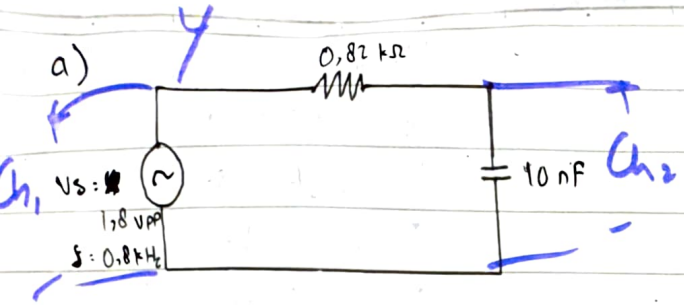
a : 0,4
c : 2,5

3. Diketahui : V/div : 0,5 V
n : 4,4
Jwb : $V_{DC} = \pm n \times V/div$
: 4,4 x 0,5 = 2,25 V

4. Diketahui : R = 0,82 kΩ = 820 Ω
C = 10 nF = 10 x 10^{-9}
V_S = 1,8 V_{PP}
f = 0,8 kHz = 800 Hz

Jawab →

a)



b) $\omega = 2\pi f = 2\pi(800) = 5026,55$

$\varphi = -\tan^{-1}(\omega RC) = -\tan^{-1}((5026,55)(820)(10\times10^{-9})) = -2,36°$

$$V_o = \frac{1}{\sqrt{1^2+\omega^2R^2C^2}} \times V_s = \frac{1}{\sqrt{1^2+(5026,55)^2\times(820)^2\times(10\times10^{-9})^2}} \times 1,8 = 1,798 \text{ Vpp} \approx 1,8 \text{ Vpp}$$

c) $V_{in\ RMS} = \frac{V_{pp}}{2\sqrt{2}} = \frac{\frac{1,8}{2}}{\sqrt{2}} = \frac{0,9}{\sqrt{2}} = 0,636 \text{ V}$

$V_{out\ RMS} = \frac{V_{pp}}{2\sqrt{2}} = \frac{\frac{1,798}{2}}{\sqrt{2}} = \frac{0,899}{\sqrt{2}} = 0,6356 \text{ V} \approx 0,636 \text{ V}$

5. a) Diketahui : I = 0,0092 A

Ditanyakan : I dalam mA ?

Jawab : $I = \frac{0,0092}{10^{-3}} = 0,0092 \times 10^3 = 9,2 \text{ mA}$

b) Diketahui : C = 47 nF = 47×10^{-9}

Ditanyakan : C dalam μF ?

Jawab : $C = \frac{47\times10^{-9}}{10^{-6}} = 47\times10^{-9} \times 10^6 = 47\times10^{-9+6} = 47\times10^{-3} = 0,047 \text{ μF}$